Candidate Name | Centre Number | Candidate Number

ZIMSEC
For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

WOOD TECHNOLOGY AND DESIGN — 6027/1
Paper 1 Theory, Graphics and Design — 3 hours

SPECIMEN PAPER

Additional materials:
Answer sheet,
Coloured crayons,
Drawing paper (**A2**),
Metric scale rule, scale of **1:1** and **1:5**,
Standard drawing equipment and
Scientific calculator.

INSTRUCTIONS TO CANDIDATES
Write your name, centre number and candidate number in the spaces at the top of this page and on **all** separate answer paper used.

INFORMATION FOR CANDIDATES

Marks are given in brackets [ ] at the end of **each** question or **part** question.
This question paper consists of **three** compulsory sections.

FOR EXAMINER'S USE	
A	
B5	
C6	
C7	
TOTAL	

This specimen paper consists of 8 printed pages.

[Turn over

For Examiner's Use

SECTION A (40 marks)

*Answer **all** questions in this section in the spaces provided.*

*You are advised to spend **not** more than **20 minutes** on this section.*

1 **(a)** Give **two** advantages and **two** disadvantages of **CAD** in Wood Technology and Design.

Advantages

1. ______________________________

____

2. ______________________________

______________________________ [2]

Disadvantages

1. ______________________________

____

2. ______________________________

______________________________ [2]

(b) In the production process briefly explain what is meant by the following terms:

(i) quality assurance,

____

______________________________ [2]

(ii) quality control.

____

______________________________ [2]

For Examiner's Use

(c) What is the purpose of Quality staff in manufacturing?

____

____

____

____ [2]

2 **(a)** Explain what is meant by the term *patent*.

____

____

____

____ [2]

(b) List down any **five** procedures for patent registration.

1. ____

2. ____

3. ____

4. ____

5. ____

[5]

(c) Give any **three** benefits of patents to a Wood Technologist.

1. ____

____

2. ____

____

3. ____

____

[3]

For Examiner's Use

3 **(a)** What is meant by the term *disaster management*?

____

____

____

____ [2]

(b) Briefly explain how we can effectively manage disasters in the workshop.

____

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____

____

____

____

____ [6]

(c) What precautions must be taken when carrying out processes that create

(i) dust,

____

____ [1]

(ii) toxic fumes.

____

____ [1]

For Examiner's Use

4 **(a)** Identify and explain **three** types of static stress to which materials can be subjected to.

(i) ____

____

____

____ [2]

(ii) ____

____

____

____ [2]

(iii) ____

____

____

____ [2]

(b) A computer desk top will be made from a manufactured board.

(i) Name a suitable manufactured board for the desk top.

____ [1]

(ii) Give a reason for choosing the board in **(b)(i)**.

____

____ [1]

(iii) Briefly explain how the manufactured board is produced.

____

____

____

____ [2]

[Turn over

SECTION B (40 marks)

*Answer all questions on **A2** plain paper provided.*

*You are advised to spend **not** more than **1 hour 50 minutes** on this section.*

5 A young couple occasionally enjoy playing table tennis in their apartment garden. However, they are facing a problem of storing the table in their small apartment because of limited space.

(a) Design a folding and detachable table to fit in the small space available in their flat. [8]

(b) By means of detailed sketches, show the following:

(i) the folding mechanism to be used on the table after use. [10]

(ii) the locking mechanism to be employed during use. [2]

(c) Show the knock-down fitting you would use to attach the table to the underframe. [4]

(d) State **three** factors which are important in designing the unit. [3]

(e) You are required to produce a working drawing in the form of:

(i) a front view,

(ii) sectional end elevation of the complete table tennis unit.

[10]

NB: Diagrams to be free hand and not to scale.

(f) Apart from wood, state **two** other materials you would use to make components of the table tennis unit. [2]

(g) State a suitable finish you would use on the unit. [1]

SECTION C (20 marks)

*Answer **one** question only from this section.*

*You are advised to spend **not** more than **40 minutes** on this section.*

6 **(a)** **(i)** A bar has to be machined to produce a sharp end as shown in **Fig. 1**.

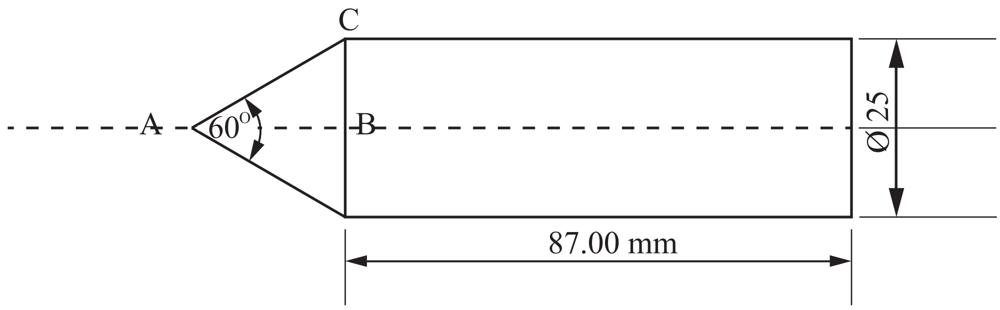

Fig. 1

Calculate the minimum length of bar required. [5]

(ii) A casting has a mass of **20 kg**. Calculate the work done in lifting it through a height of **8 m**. [5]

(b) A newly established business enterprise has the following information:

Sales of the year	30 000 units
Direct cost per unit:	
Labour	$200
Materials	$175
Fixed costs	$2 000 000
Selling price per unit	$600

(i) Calculate:

1. revenue cost, [3]

2. total cost, [3]

3. profit. [1]

(ii) Calculate the profit given that sales decreased by 20% and selling price increased by 15%. [3]

[Turn over

7 **Fig. 2** shows a roofing truss for a two roomed classroom block is to be constructed using angle irons of **50 × 50 × 5 mm** for the rafters, tie-beams, king posts and queen posts.

All struts are to be made by an angle iron of **30 × 30 × 5 mm**. The king post is to be **1.8 m** long, whilst the queen posts labelled **A** and those labelled **B** are to be **1.2 m** and **0.9 m** long respectively.

Also, the struts labelled **C** and **D** are to be **1.5 m** and **1.2 m** long respectively.

Given that the width of the block is **8.2 m** wide with overlapping verandahs on the tie-beam to be **0.8 m** and **0.4 m** respectively at front and back.

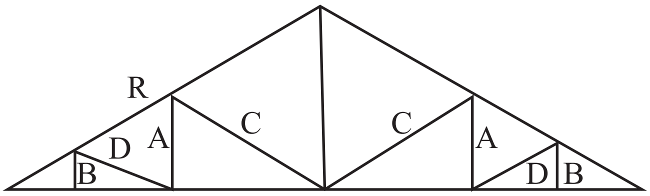

Fig. 2

(a) Find

(i) the length of the tie-beam, [2]

(ii) the length of the rafter labelled **R**, [2]

(iii) the number of angle irons (**50 × 50 × 5 mm**) full lengths **6 m** long each required to construct the rafters, tie beams, king posts and queen posts. [5]

(iv) the number of irons **30 m × 30 m × 5 mm** full lengths **6 m** long each) required to complete the struts. [5]

(b) The length of the classroom block to be roofed is **20.1 m** long and spacing between trusses is **2.1 m**.

Determine the number of trusses required for the full block. [6]